Node.js Server Deployment on AWS EC2 using PM2

AWS EC2 Node.js 18 PM2 Process Manager

Project Overview

This project demonstrates deployment of a Node.js application on AWS EC2 using:

- AWS EC2
- Node.js
- Express.js
- PM2
- GitHub
- Amazon Linux 2023

The application is publicly accessible using the EC2 Public IP on port `3000`.

Architecture

```
User Browser
      |
      ▼
AWS EC2 Instance
      |
      ▼
Node.js Application
      |
      ▼
PM2 Process Manager
```

Project Structure

```
node-server-project/
│
├── app.js
├── package.json
├── README.md
└── screenshots/
```

AWS EC2 Details

Service	Details
Cloud Provider	AWS
Instance Type	t3.micro
Operating System	Amazon Linux 2023
Runtime	Node.js 18
Process Manager	PM2

Step 1 — Launch EC2 Instance

1. Open AWS Console
2. Go to EC2 Dashboard
3. Click **Launch Instance**
4. Select:
   - Amazon Linux 2023
   - t3.micro
5. Configure Security Group:
   - SSH → Port 22
   - HTTP → Port 80
   - Custom TCP → Port 3000

Step 2 — Connect to EC2

```
ssh -i your-key.pem ec2-user@YOUR_PUBLIC_IP
```

Step 3 — Update Server

```
sudo yum update -y
```

Step 4 — Install Node.js

```
sudo yum install nodejs -y
```

Verify Installation

```
node -v
npm -v
```

Step 5 — Clone GitHub Repository

```
git clone https://github.com/YOUR_USERNAME/node-server-project.git
```

Move into Project Directory

```
cd node-server-project
```

Step 6 — Install Dependencies

```
npm install
```

Step 7 — Install PM2

```
sudo npm install pm2 -g
```

Step 8 — Start Application using PM2

```
pm2 start app.js
```

Enable Auto Restart on Server Reboot

```
pm2 startup
pm2 save
```

Check Running Applications

```
pm2 list
```

Access Application

Open browser:

```
http://YOUR_PUBLIC_IP:3000
```

Example:

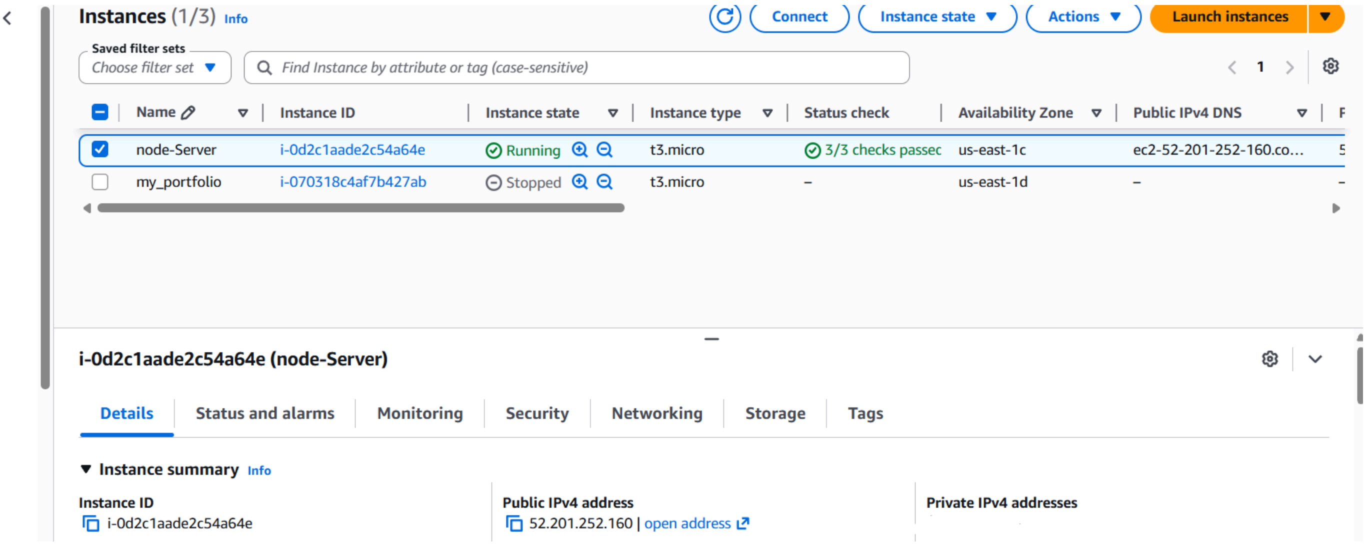
```
http://52.201.252.160:3000
```

Application Output

```
Node Server
Hello from AWS EC2 + Node.js + PM2
CI/CD Deployment Successful
```

Screenshots

AWS EC2 Dashboard



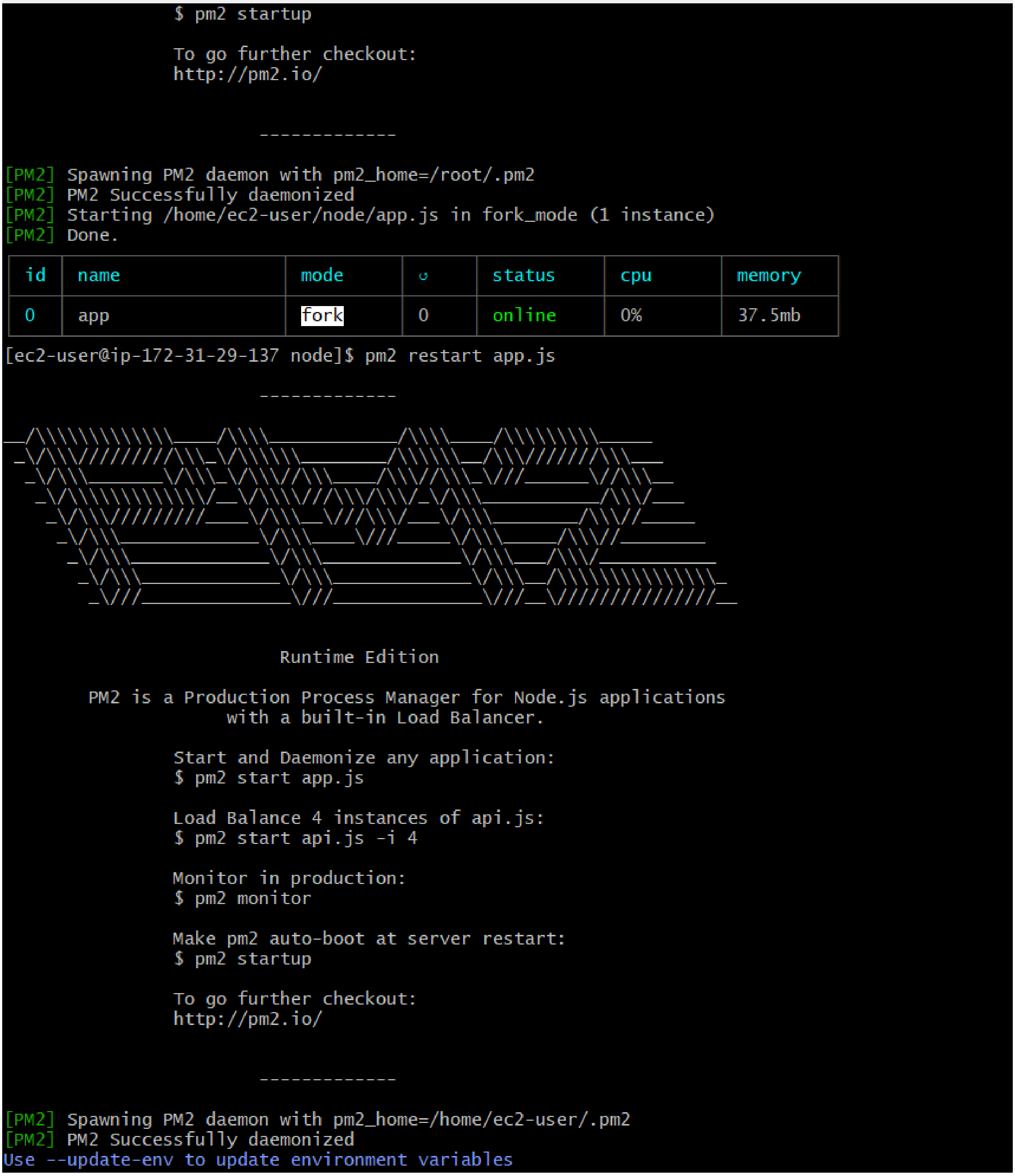
Install Node.js

```
[ec2-user@ip-172-31-29-137 ~]$ sudo yum update
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.
Nothing to do.
Complete!
[ec2-user@ip-172-31-29-137 ~]$ sudo yum install nodejs -y
Last metadata expiration check: 0:00:19 ago on Sat May 16 07:36:58 2026.
Dependencies resolved.
================================================================================================
 Package                                   Architecture                   Version
================================================================================================
Installing:
 nodejs                                    x86_64                         1:18.20.8-1.amzn2023.0.2
Installing dependencies:
 libbrotli                                 x86_64                         1.0.9-4.amzn2023.0.2
 nodejs-libs                               x86_64                         1:18.20.8-1.amzn2023.0.2
Installing weak dependencies:
 nodejs-docs                               noarch                         1:18.20.8-1.amzn2023.0.2
 nodejs-full-i18n                          x86_64                         1:18.20.8-1.amzn2023.0.2
 nodejs-npm                                x86_64                         1:10.8.2-1.18.20.8.1.amzn2023

Transaction Summary
================================================================================================
Install  6 Packages

Total download size: 45 M
Installed size: 224 M
Downloading Packages:
(1/6): libbrotli-1.0.9-4.amzn2023.0.2.x86_64.rpm
(2/6): nodejs-docs-18.20.8-1.amzn2023.0.2.noarch.rpm
(3/6): nodejs-full-i18n-18.20.8-1.amzn2023.0.2.x86_64.rpm
(4/6): nodejs-18.20.8-1.amzn2023.0.2.x86_64.rpm
(5/6): nodejs-npm-10.8.2-1.18.20.8.1.amzn2023.0.2.x86_64.rpm
(6/6): nodejs-libs-18.20.8-1.amzn2023.0.2.x86_64.rpm
------------------------------------------------------------------------------------------------
Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing        :
  Installing       : nodejs-docs-1:18.20.8-1.amzn2023.0.2.noarch
  Installing       : libbrotli-1.0.9-4.amzn2023.0.2.x86_64
  Installing       : nodejs-libs-1:18.20.8-1.amzn2023.0.2.x86_64
  Installing       : nodejs-full-i18n-1:18.20.8-1.amzn2023.0.2.x86_64
  Installing       : nodejs-npm-1:10.8.2-1.18.20.8.1.amzn2023.0.2.x86_64
  Installing       : nodejs-1:18.20.8-1.amzn2023.0.2.x86_64
  Running scriptlet: nodejs-1:18.20.8-1.amzn2023.0.2.x86_64
INFO: registered node-18 in the alternatives

  Verifying        : libbrotli-1.0.9-4.amzn2023.0.2.x86_64
  Verifying        : nodejs-1:18.20.8-1.amzn2023.0.2.x86_64
  Verifying        : nodejs-docs-1:18.20.8-1.amzn2023.0.2.noarch
  Verifying        : nodejs-full-i18n-1:18.20.8-1.amzn2023.0.2.x86_64
  Verifying        : nodejs-libs-1:18.20.8-1.amzn2023.0.2.x86_64
  Verifying        : nodejs-npm-1:10.8.2-1.18.20.8.1.amzn2023.0.2.x86_64
```

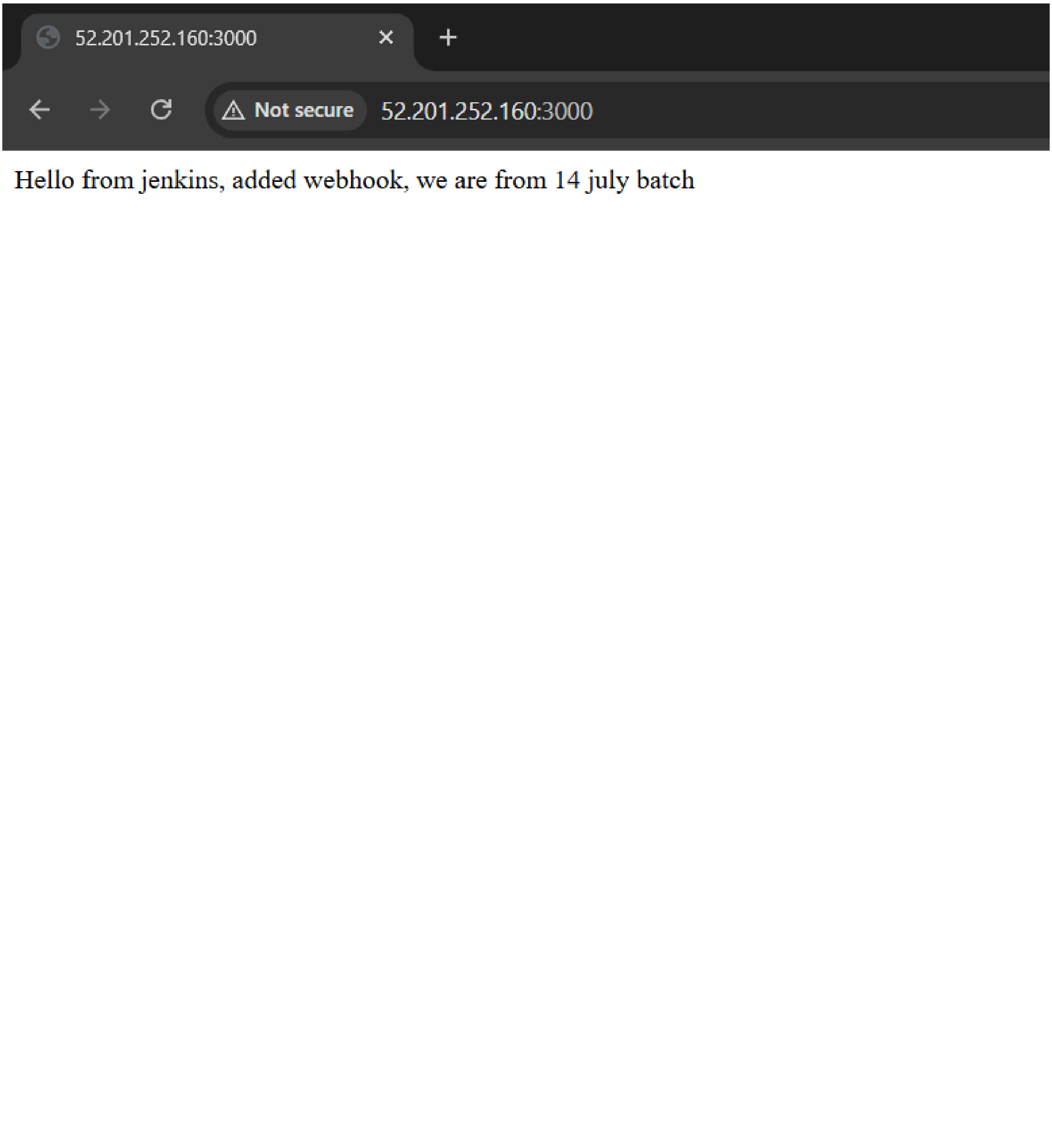
GitHub Clone & NPM Install

```
Complete!
[ec2-user@ip-172-31-29-137 ~]$ git clone https://github.com/iamtruptimane/node-js-app-CICD.git
Cloning into 'node-js-app-CICD'...
remote: Enumerating objects: 49, done.
remote: Counting objects: 100% (31/31), done.
remote: Compressing objects: 100% (18/18), done.
remote: Total 49 (delta 20), reused 21 (delta 13), pack-reused 18 (from 1)
Receiving objects: 100% (49/49), 5.54 KiB | 5.54 MiB/s, done.
Resolving deltas: 100% (24/24), done.
[ec2-user@ip-172-31-29-137 ~]$ ls
node-js-app-CICD
[ec2-user@ip-172-31-29-137 ~]$ mv node-js-app-CICD/ node
[ec2-user@ip-172-31-29-137 ~]$ ls
node
[ec2-user@ip-172-31-29-137 ~]$ cd/ node
-bash: cd/: No such file or directory
[ec2-user@ip-172-31-29-137 ~]$ cd /node
-bash: cd: /node: No such file or directory
[ec2-user@ip-172-31-29-137 ~]$ cd node
[ec2-user@ip-172-31-29-137 node]$ sudo npm install
```

PM2 Running Application



Final Browser Output



Useful PM2 Commands

Restart Application

```
pm2 restart app.js
```

Stop Application

```
pm2 stop app.js
```

Delete Application

```
pm2 delete app
```

View Logs

```
pm2 logs
```

Security Group Configuration

Type	Port
SSH	22
HTTP	80
Custom TCP	3000

Future Improvements

- Jenkins CI/CD Pipeline
- Nginx Reverse Proxy
- Docker Containerization
- HTTPS SSL Setup
- Application Load Balancer
- Auto Scaling Group
- Route53 Domain Configuration

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summary

Successfully deployed a Node.js application on AWS EC2 using PM2 and GitHub integration. The server is publicly accessible and production-ready for future CI/CD implementations.
